

Portfolio Analysis Project Assignment

Introduction

The objective of this project was to create an equal-weighted portfolio of seven publicly traded stocks and compare its risk and return characteristics against three major market ETFs: SPY, IWM, and DIA.

The portfolio required calculating individual stock factors, comparing each stock and the full portfolio against SPY, IWM, and DIA, and producing a correlation matrix between the portfolio, ETFs, and individual assets.

Data

The portfolio consists of the following seven publicly traded stocks:

Stock	Company Name	Industry
T	AT&T Inc.	Diversified Telecommunication Services
ORCL	Oracle Corporation	Software
DAL	Delta Air Lines, Inc.	Passenger Airlines
BBY	Best Buy Co., Inc.	Specialty Retail
DIS	The Walt Disney Company	Entertainment
CAT	Caterpillar Inc.	Machinery
WRB	W. R. Berkely Corporation	Insurance

The idea was to build a somewhat diversified portfolio across technology, media, retail, airlines, industrials, and insurance.

Daily close prices for the seven selected stocks and the three ETFs (SPY, IWM, and DIA) from 2016-03-24 to 2026-03-24 were downloaded using *yfinance*, then converted into daily returns for risk and performance calculations.

Methodology

The analysis began by downloading 10 years of closing prices for the seven selected stocks and the three ETFs.

There are three parts to this project:

Individual stocks vs each ETF

The first table analyzes the seven individual stocks in the portfolio.

Portfolio Weight	Portfolio returns were calculated using a constant equal-weighted approach, meaning each stock contributes 14.29% to the portfolio return calculation throughout the analysis period.
Annualized volatility	Annualized volatility was calculated using the trailing 3 months of daily returns. The daily standard deviation was multiplied by square root of 252 (252 for number of trading days) - Most volatile stock – ORCL (51.14%) - Least Volatile – T and WRB (25.65% and 20.38% respectively)
Beta Against SPY, IWM, and DIA	Beta was calculated as the covariance between the stock's returns and the market ETF's returns divided by the variance of the market ETF's returns. T is the most defensive holding based on beta, with beta near zero against SPY and IWM and only 0.20 against DIA. WRB is also defensive, with beta below 0.40 against all three ETFs. By contrast, DAL is the most aggressive holding, with beta above 1.65 against all benchmarks and above 2.0 against DIA.
Maximum Weekly Drawdown	DAL had the highest weekly drawdown of 68.85%. WRB had the smallest weekly drawdown, around 41.83%, making it the most stable by this measure.
Average Weekly Drawdown	Average weekly drawdowns ranged from 25.00% to 38.23%. T had the least severe average drawdown, while DAL and BBY had the most severe drawdowns.
Total Return and Annualized Return	CAT had the highest total return of about 1103.87%, and an annualized return of about 28.25%. Similarly, WRB also showed a strong total return of about 411.51%, with an annualized return of about 17.73%. DIS performed the worst in both metrics, with a total return of only 7.56% and an annualized return of about 0.7%.

Portfolio vs each ETF.

Correlations	Our portfolio showed high positive correlations with all three ETFs (between 0.82 to 0.87). Hence, the portfolio moved in the same direction as the broad market ETFs.
Covariance	Ranging between 0.000126 to 0.000158, the covariance values were small but positive for all three ETFs, meaning the portfolio tended to move in the same direction as market ETFs.
Tracking Error	It measures how much the portfolio's returns differs from an ETF's Return. The portfolio had tracking errors between 10% to 13% against the ETFs. This indicates that although the portfolio is

highly correlated with the ETFs, it still has meaningful differences from them. This makes sense because the portfolio contains only seven stocks compared to substantially larger and more diversified baskets of securities that the ETFs consist of.

Portfolio Sharpe Ratio	The Sharpe ratio measures the risk-adjusted return. The risk-free rate was assumed to be 3% annually. The portfolio Sharpe ratio was approximately 0.632. This means the portfolio generated about 0.632 units of excess return per unit of risk.
ETF Sharpe Ratio	Since the portfolio itself has only one Sharpe ratio, I also calculated the Sharpe ratio for each ETF to make the risk-adjusted comparison clearer.
Sharpe Difference	Compared to SPY, the portfolio had a slightly lower Sharpe ratio (-0.036), meaning that Spy had slightly better risk-adjusted performance than the portfolio. At 0.224 higher value than that of IWM's Sharpe ratio, the portfolio performed better than IWM on a risk-adjusted basis. Portfolio only slightly outperformed DIA on a risk-adjusted basis (0.046). Overall, portfolio's Sharpe ratio was better than IWM and DIA, but slightly worse than SPY.
Annualized Volatility Spread	Calculated as the difference of ETF volatility from Portfolio volatility, annualized volatility measures whether the portfolio is more or less volatile than the ETFs. The volatility spread was the smallest against IWM, meaning the portfolio's volatility was closest to the Russell 2000 ETF. The spread was larger against SPY and against DIA, meaning the portfolio was noticeably more volatile than those benchmarks.

Chart Interpretations

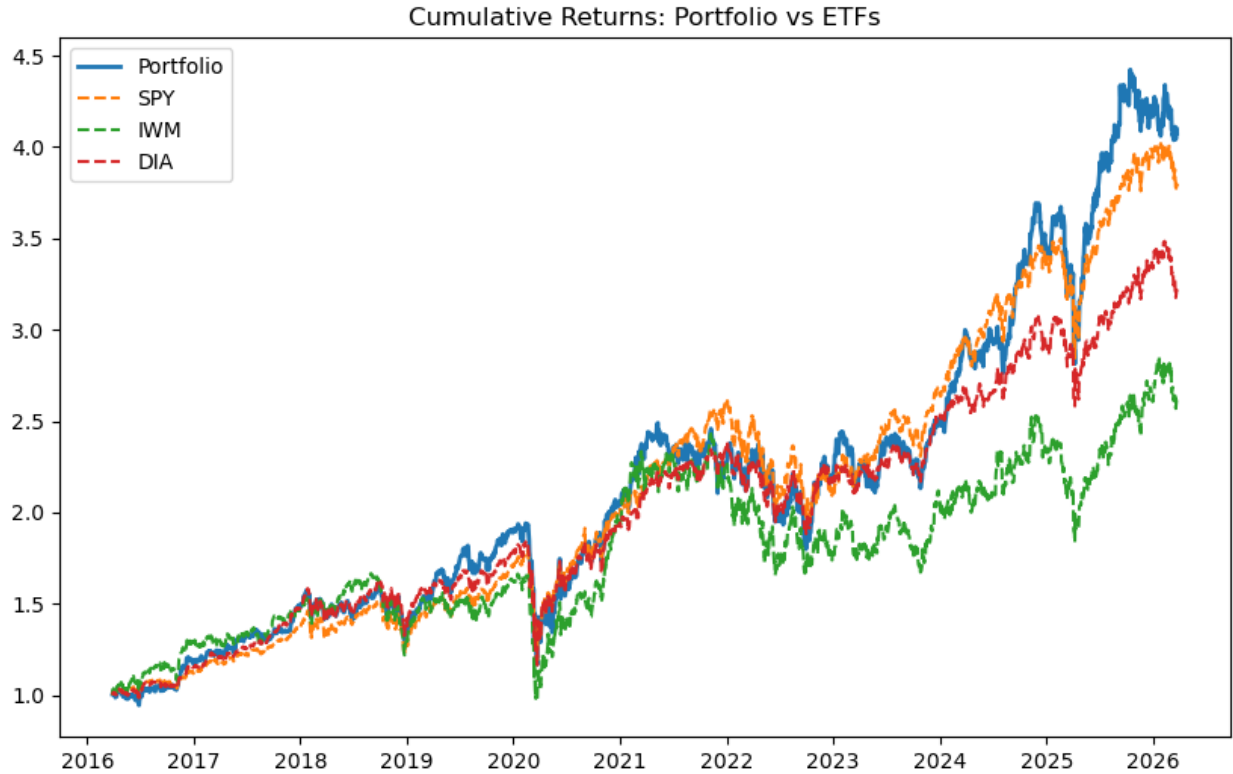


Figure 1 - Cumulative Returns: Portfolio vs ETFs

Figure 1 shows the comparison between the growth of the equal-weighted portfolio against DIA, IWM, and SPY over the full sample period. The portfolio generally moved in the same direction as the ETFs, which matches the high correlations from the portfolio comparison table. The portfolio produced stronger long-term growth than the benchmark ETFs. The chart also highlights that our portfolio had larger swings especially on the upside. The portfolio outperformed all 3 ETFs over time.

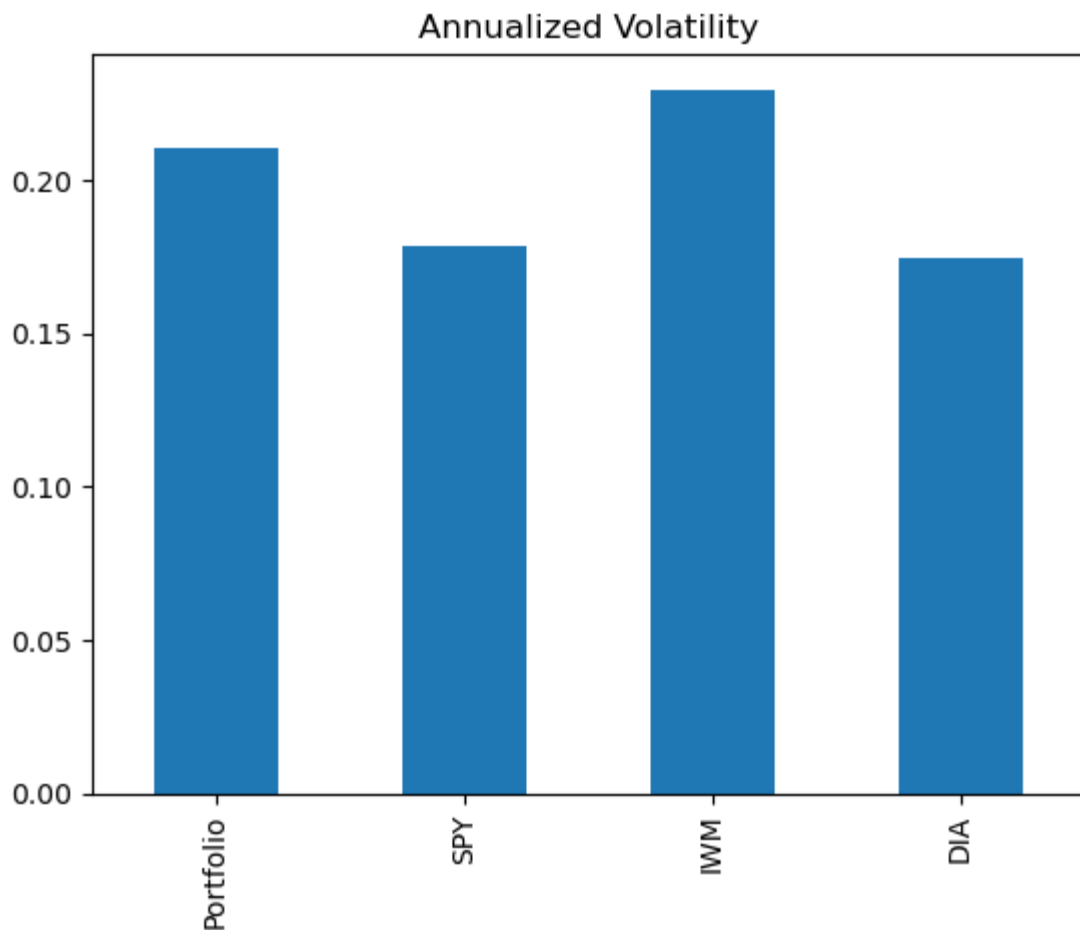


Figure 2 - Annualized Volatility Chart

Figure 2 compares the annualized volatility of the portfolio against DIA, IWM, and SPY. It shows that the portfolio was more volatile than SPY and DIA, but less volatile than IWM. The portfolio's risk level was closer to the Russel 2000 ETF than to the S&P 500 or Dow Jones ETFs. Since the portfolio only contains seven stocks, it is expected to have more concentrated risk than broad ETFs.

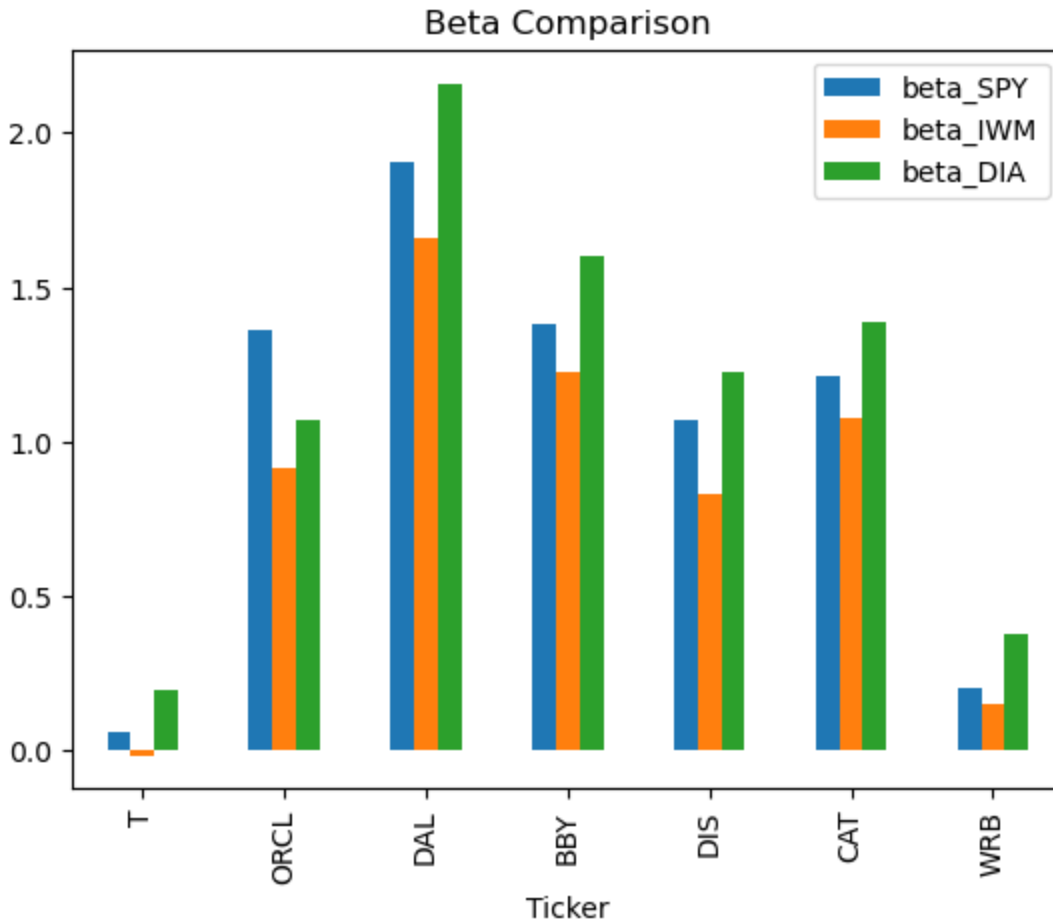


Figure 3 - Beta Comparison

Figure 3 visually confirms the beta results from the first table. DAL had the highest beta across ETFs, making it the most market-sensitive stock in the portfolio. Its beta exceeded 1.65 against all three benchmarks and was above 2.0 against DIA, indicating that it tended to amplify market movements. On the other hand, T had the lowest beta values, with betas close to zero against SPY and IWM and only 0.20 against DIA, making it the most defensive holding. WRB was also relatively defensive, with beta values below 0.40 across all three ETFs.

Correlation matrix of individual stocks, portfolio consisting of the individual stocks, and three ETFs among each other.

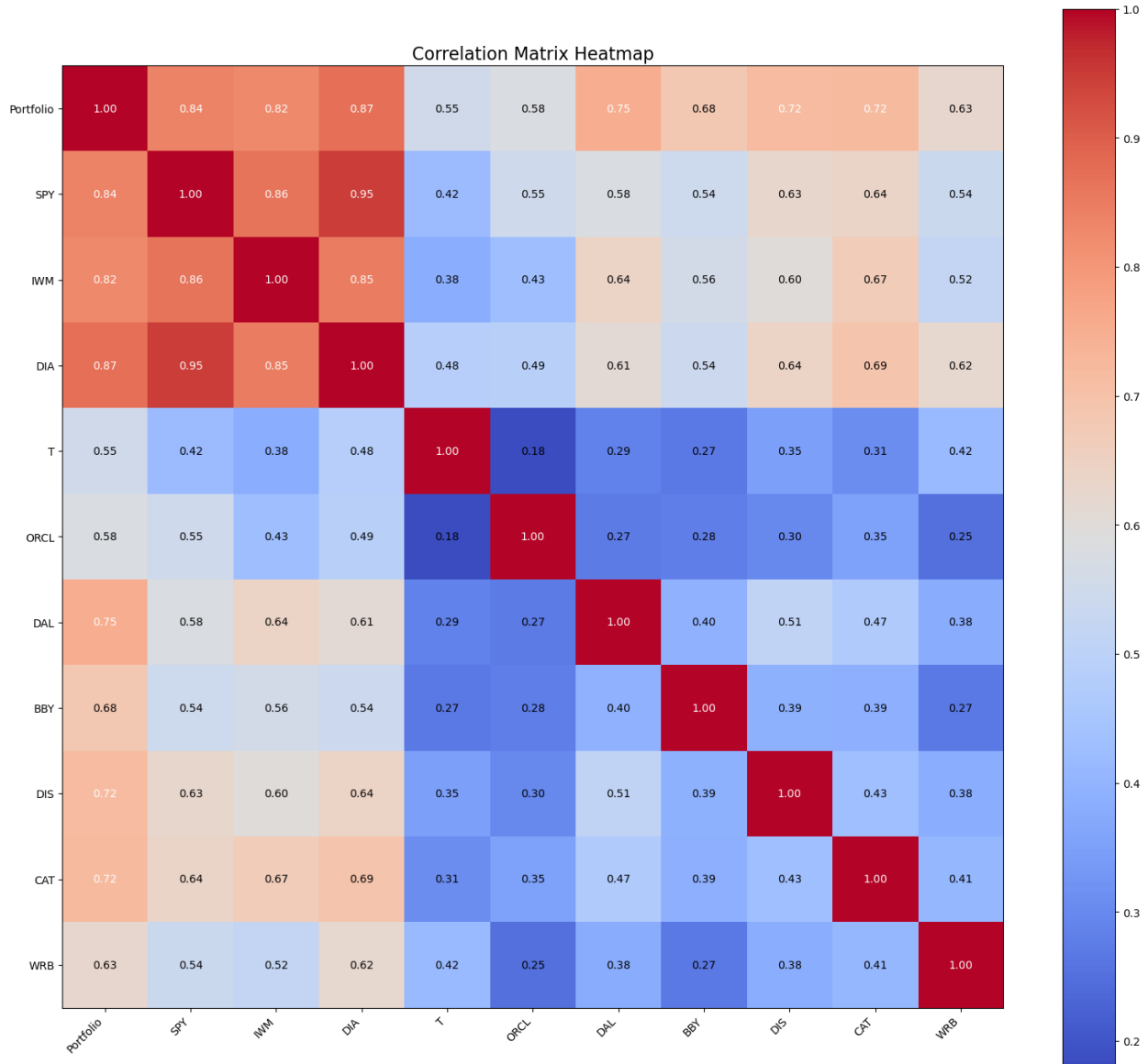


Figure 4 - Correlation Matrix Heatmap

Figure 4 shows how closely each asset moves with every other asset. The portfolio has high correlation with all 3 ETFs. Portfolio vs DIA is 0.87, Portfolio vs IWM is 0.82, and Portfolio vs SPY is 0.84.

The 3 ETFs are highly correlated with each other with their values ranging from 0.85 to 0.95. SPY and DIA (0.95) are especially close because both are large-cap US equity benchmarks.

The individual stocks are less correlated with each other ranging from 0.16 to 0.70. Many of the stock-to-stock correlations are lower than the ETF-to-ETF correlations, which suggests that the individual stocks provide some diversification benefits. However, the portfolio is

still highly correlated with the ETFs, so it remains strongly exposed to broad market movements.

T has the lowest correlation with the overall portfolio, around 0.54, which supports the idea that WRB acts as the most defensive holding in the portfolio. Overall, the heatmap shows that the portfolio is strongly connected to the market, but the individual stock correlations are not so high that diversification disappears.

Conclusion

This project analyzed an equal-weighted portfolio consisting of seven NYSE-listed stocks and compared its performance against SPY, IWM, DIA over a ten-year period.

The portfolio moves strongly with the overall market, especially SPY and DIA. The portfolio had high positive correlations with all three ETFs, which shows that it generally followed market direction. With only seven selected stocks, the portfolio presented a more concentrated risk. Among the benchmarks, the portfolio was most closely related to DIA, indicating that its behavior was most similar to large-cap value and industrial-oriented companies.

Individual stock analysis showed substantial differences in risk and performance characteristics. CAT was the strongest long-term performer, generating the highest total return and annualized return among the selected stocks. WRB and T served as the most defensive holdings, exhibiting relatively low volatility and low beta values. In contrast, DAL displayed the highest market sensitivity and experienced the largest drawdowns during periods of market stress.

From a portfolio perspective, the portfolio achieved higher cumulative growth than SPY, IWM, and DIA over the sample period. Although it carried more volatility than SPY and DIA, it delivered stronger long-term returns. The portfolio's Sharpe ratio exceeded those of IWM and DIA, while remaining slightly below that of SPY, indicating competitive risk-adjusted performance.

Overall, the portfolio benefited from diversification across multiple industries while maintaining strong market exposure. The results suggest that a concentrated but diversified selection of individual stocks can outperform broad market benchmarks over time, although investors must be willing to accept higher volatility and larger drawdowns in exchange for the potential for higher returns.